

Supplementary Table 4: Sex differences in cross-trait genetic correlations

Pair	rg(se) males	rg(se) fe- males	P diff adj
HDL—LDL	0.1617 (0.0236)	-0.1129 (0.0237)	0.00
LDL—Waist circumference	-0.1425 (0.0312)	0.1849 (0.028)	0.00
Educational attainment—Risk-taking behaviour	-0.1431 (0.034)	0.1881 (0.0335)	0.00
Body Mass Index—LDL	-0.0746 (0.0246)	0.1463 (0.0232)	0.00
LDL—nonHDL	0.8851 (0.0113)	0.9604 (0.0036)	0.00
LDL—logTG	0.1271 (0.0319)	0.3626 (0.024)	0.00
LDL—Waist-hip ratio	-0.007 (0.0356)	0.2811 (0.0339)	0.00
LDL—Weight	-0.1567 (0.0312)	0.0887 (0.0319)	0.00
Waist circumference (adjBMI)—Waist- hip ratio	0.3631 (0.0392)	0.6303 (0.0284)	0.00
TC—Waist circumference	-0.1675 (0.0309)	0.0531 (0.0267)	0.00
nonHDL—Waist circumference	0.0201 (0.036)	0.2638 (0.0294)	0.00
LDL—Type 2 diabetes	-0.0146 (0.0306)	0.187 (0.0263)	0.00
Hip circumference (adjBMI)—Waist circumference (adjBMI)	0.6183 (0.0301)	0.3468 (0.0466)	0.00
Body Mass Index—Waist-hip ratio	0.6636 (0.0279)	0.443 (0.0359)	0.00
Body Mass Index—Weight	0.7817 (0.0185)	0.8869 (0.0134)	0.00
Waist-hip ratio—Waist-hip ratio (ad- jBMI)	0.7266 (0.025)	0.8568 (0.0137)	0.00
Hip circumference—LDL	-0.152 (0.0356)	0.0495 (0.0272)	0.00
LDL—Waist circumference (adjBMI)	-0.0915 (0.0331)	0.123 (0.035)	0.00
TC—Weight	-0.2007 (0.0314)	-0.0116 (0.0298)	0.00
Age at first birth—Risk-taking be- haviour	-0.3394 (0.0542)	-0.035 (0.0463)	0.00
Fasting insulin—LDL	-0.1201 (0.0538)	0.2207 (0.0592)	0.00
Waist circumference (adjBMI)—Waist- hip ratio (adjBMI)	0.5319 (0.0364)	0.717 (0.0243)	0.00
nonHDL—Waist circumference (ad- jBMI)	-0.0364 (0.0331)	0.1627 (0.0348)	0.00
HDL—nonHDL	-0.118 (0.0291)	-0.2753 (0.0248)	0.00

Body Mass Index—TC	-0.0975 (0.0238)	0.0318 (0.0211)	0.00
Body Mass Index—nonHDL	0.0638 (0.0275)	0.2178 (0.0261)	0.00
logTG—Waist circumference (adjBMI)	0.1024 (0.0361)	0.2879 (0.0299)	0.00
Height—Weight	0.5836 (0.0289)	0.4191 (0.0308)	0.00
Risk-taking behaviour—Schizophrenia	0.1601 (0.0322)	0.3543 (0.0388)	0.00
Insomnia—Schizophrenia	0.1442 (0.0356)	-0.0532 (0.0375)	0.00
nonHDL—Weight	-0.0444 (0.034)	0.1387 (0.0339)	0.00
LDL—Waist-hip ratio (> 50 yrs, adjBMI)	0.0118 (0.0465)	0.2463 (0.0418)	0.00
nonHDL—Waist-hip ratio	0.1792 (0.0435)	0.3853 (0.0337)	0.00
Hip circumference—TC	-0.2028 (0.0347)	-0.0449 (0.0242)	0.00
Hip circumference—Waist circumference (adjBMI)	0.3285 (0.0433)	0.1218 (0.0356)	0.00
Free thyroxine (FT4)—Waist-hip ratio (adjBMI)	0.1878 (0.0767)	-0.1965 (0.0747)	0.00
HDL—Waist circumference (adjBMI)	-0.1276 (0.031)	-0.2788 (0.0284)	0.00
HDL—Height SDS	-0.19 (0.05)	0.0362 (0.0387)	0.00
Waist circumference (adjBMI)—Waist-hip ratio (> 50 yrs, adjBMI)	0.5403 (0.0435)	0.7287 (0.0313)	0.01
Body Mass Index—Hip circumference	0.7946 (0.0204)	0.8768 (0.0116)	0.01
ADHD—Schizophrenia	0.1851 (0.0414)	-0.1201 (0.0787)	0.01
Height—Hip circumference	0.4594 (0.0375)	0.299 (0.029)	0.01
Educational attainment—Insomnia	-0.1998 (0.0427)	-0.3746 (0.0301)	0.01
Type 2 diabetes—Waist circumference (adjBMI)	0.0858 (0.0426)	0.2778 (0.0391)	0.01
Fasting glucose—LDL	-0.1275 (0.0492)	0.101 (0.0487)	0.01
Waist circumference (adjBMI)—Waist-hip ratio (<= 50 yrs, adjBMI)	0.5814 (0.0617)	0.8404 (0.0486)	0.01
TC—Waist-hip ratio	-0.007 (0.0366)	0.1523 (0.0319)	0.01
LDL—Waist-hip ratio (adjBMI)	0.0546 (0.0407)	0.2281 (0.0352)	0.01
Fasting insulin—Waist-hip ratio (> 50 yrs, adjBMI)	0.1185 (0.0871)	0.5428 (0.099)	0.01
Educational attainment—Multisite chronic pain	-0.5404 (0.029)	-0.4186 (0.0247)	0.01

Schizophrenia—Snoring	-0.2013 (0.0312)	-0.0599 (0.032)	0.01
Number of children ever born—Risk-taking behaviour	0.4151 (0.0724)	0.1117 (0.0639)	0.01
Cardiovascular disease—Waist circumference (adjBMI)	0.0023 (0.0367)	0.1873 (0.0465)	0.02
Hip circumference—nonHDL	-0.0675 (0.0397)	0.091 (0.0319)	0.02
TC—Waist circumference (adjBMI)	-0.0837 (0.0298)	0.0579 (0.0343)	0.02
Fasting insulin—TC	-0.1203 (0.0496)	0.1058 (0.0535)	0.02
Height SDS—Weight	0.7621 (0.0866)	0.4639 (0.0498)	0.02
Grip strength measurement/ Muscle weakness—Left 5m area	-0.1645 (0.1186)	0.2555 (0.0787)	0.03
Height—Waist-hip ratio (adjBMI)	-0.1434 (0.0498)	0.0384 (0.0379)	0.03
Height—Hip circumference (adjBMI)	0.8467 (0.0251)	0.7502 (0.0223)	0.03
nonHDL—Waist-hip ratio (> 50 yrs, adjBMI)	0.135 (0.0494)	0.3173 (0.0399)	0.03
logTG—nonHDL	0.4832 (0.0256)	0.5759 (0.0199)	0.03
nonHDL—Type 2 diabetes	0.1913 (0.0304)	0.3085 (0.0278)	0.03
Schizophrenia—Waist-hip ratio (> 50 yrs, adjBMI)	0.0973 (0.039)	-0.0769 (0.0479)	0.03
ADHD—Age at first birth	-0.4602 (0.0638)	-0.7753 (0.0923)	0.04
Height—Waist-hip ratio (> 50 yrs, adjBMI)	-0.1014 (0.0581)	0.0967 (0.0404)	0.04
HDL—Waist-hip ratio (adjBMI)	-0.2235 (0.0425)	-0.37 (0.031)	0.04
Height—Waist-hip ratio	-0.1689 (0.0433)	-0.0122 (0.0358)	0.04
Educational attainment—Waist-hip ratio (> 50 yrs, adjBMI)	-0.2743 (0.0443)	-0.1089 (0.0398)	0.04
logTG—Waist-hip ratio (adjBMI)	0.3143 (0.038)	0.4509 (0.0315)	0.04
HDL—Snoring	-0.1587 (0.0262)	-0.2537 (0.0226)	0.04
Height SDS—Hip circumference	0.6936 (0.0904)	0.4014 (0.0589)	0.04
logTG—Waist-hip ratio	0.4628 (0.0374)	0.5927 (0.0305)	0.04